Joint $(\boldsymbol{c}, \pi)$ Bellman DP vs. PCRB Baseline for the Superconducting-Qubit Flux Sensor

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Abstract

We numerically realize the joint geometry–policy framework of `AdaptiveDesign.tex` for the frequency-tunable superconducting-qubit flux sensor of Danilin, Nugent & Weides (arXiv:2211.08344v4, 2024). The unknown state $x = \Phi_{\text{ext}}$ is continuous; the sensor control $\boldsymbol{s} = (\tau, n)$ is a discrete $(J \times L)$ action; the geometry $\boldsymbol{c}$ is continuous and 7-dimensional (tier 1 of `scqubit_model.tex`). We implement the full pipeline: forward-model primitives (Phase 1), belief grid (Phase 2), enumeration baselines $V_{\text{oracle}}, V_{\text{fixed}}$ (Phase 3), exact memoized Bellman DP (Phase 4), envelope-theorem $\partial V_{\text{adaptive}}/\partial \boldsymbol{c}$ pathwise gradient (Phase 5), joint-DP outer Adam with periodic policy iteration (Phase 6), and the PCRB-baseline joint optimization (Phase 7). All gradient checks pass to machine-limited tolerance: relative error $\leq 2 \times 10^{-4}$ for the joint-DP envelope gradient (MC-noise floor) and $\leq 1.1 \times 10^{-7}$ for $\partial \log J_P / \partial \boldsymbol{c}$ (deterministic, no MC). The within-family $f_q^{\max}$ sweep reproduces the paper's optimal sensing flux $\Phi^\star/\Phi_0 = 0.4436$ at $f_q^{\max} = 9$ GHz (paper: 0.442), while exposing a qualitative divergence between frameworks: the Φ-framework V_{adaptive} peaks at $f_q^{\max} = 5$ GHz (with a broad 4–9 GHz plateau), whereas the PCRB framework's $\log J_P^\star$ grows monotonically through 20 GHz at this (K, J, L, K_Φ). The headline MSE-anchored comparison $\overline{\text{MSE}}_2/\overline{\text{MSE}}_1$ operationalizes §8.7 of the main document for this case study.

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1 Purpose and scope

`AdaptiveDesign.tex` formulates the joint geometry–policy design problem as a bilevel optimization in which the *geometry* $\boldsymbol{c}$ (fixed at runtime) is the outer variable and the *sensor policy* π (chosen online) is the inner one. Three operational questions follow.

1. **Tractability.** Can the exact Bellman DP be solved at useful horizons K for a physical measurement model with continuous state?
2. **Gradient correctness.** Does the envelope-theorem identity $\partial V_{\text{adaptive}}(\boldsymbol{c};\pi^\star)/\partial\boldsymbol{c}$ give correct, low-variance geometry gradients under a *fixed* optimal policy?
3. **Value.** In an MSE-anchored deployment comparison, does the joint-DP design $(\boldsymbol{c}_1^\star,\pi_1^\star)$ beat the PCRB-baseline design $(\boldsymbol{c}_2^\star,\boldsymbol{s}_2^\star)$ on the actual quantity of interest, $\overline{\text{MSE}} = \mathbb{E}_{\Phi,\boldsymbol{y}}[(\hat{\Phi}-\Phi)^2]$?

The companion document `IG_numerics.tex` answered (1) and (2) for a *discrete-state* radar POMDP with closed-form enumeration. This note answers them for a *continuous-state* problem grounded in actual experimental physics, and adds (3) as the §8.7 headline operationalized. The result is a worked example of path (2c) of the relaxation hierarchy (`AdaptiveDesign.tex` §10), namely: continuous x, exact Bellman DP on a K_Φ-grid, pathwise envelope-theorem gradient, and full Adam on $\boldsymbol{c}$.

2 Problem and numerical setup

Forward model. See `scqubit_model.tex` for the physical derivation. The Ramsey likelihood is

$$P_{|1\rangle}(\Phi,\tau;\,\boldsymbol{c}) \;=\; \tfrac{1}{2}+\tfrac{1}{2}\,\mathrm{th}(\Phi;\boldsymbol{c})\,e^{-A(\Phi;\boldsymbol{c})\tau-B(\Phi;\boldsymbol{c})^2\tau^2}\,\cos\!\big(\Delta\omega(\Phi;\boldsymbol{c})\,\tau\big), \tag{1}$$

with all rates $(\Gamma_{1,\text{cav}},\Gamma_{1,\text{ind}},\Gamma_{1,\text{cap}})$ and envelope coefficients $(A,\,B)$ closed-form in $\boldsymbol{c} = (f_q^{\max}, E_C/h, \kappa, \Delta_{qr}, T, A_\Phi$

Grid and horizon. Defaults used in every numerical run below:

K-grid of Φ on $[0, 0.49\Phi_0]$	$K_\Phi = 64$ (mid-point rule)
horizon	$K = 3$ decision epochs
τ-grid	(20, 40, 80, 160) ns, Kitaev ladder ($J = 4$)
n-grid	(1, 10) repetitions ($L = 2$)
baseline $\boldsymbol{c}_0$	Danilin et al. values (§3, `scqubit_plan.md`)

These defaults are deliberately modest; the code is designed to scale to plan defaults ($K = 4$, $K_\Phi = 256$, $J = 6$, $L = 4$) given parallel V-oracle / V-fixed workers (not yet wired in this run).

Module layout. The code lives in `doc/adaptive/scqubit/`:

`ScqubitModel.jl`	forward model + analytic derivatives (Pha
`Belief.jl`	grid belief, count-statistic dual rep (Phase
`Baselines.jl`	V_{oracle}, V_{fixed} enumeration (Phase 3)
`Bellman.jl`	memoized backward induction on counts (
`Gradient.jl`	exact + MC envelope-theorem gradient (P
`JointOpt.jl`	outer Adam + policy iteration (Phase 6)
`PCRB.jl`	Fisher-info, J_P, PCRB baseline, deployed
`sweep_{fq,joint,pcrb}.jl`, `compare_mse.jl`, `plot_results.jl`	drivers

3 Forward-model validation (Phase 1)

All Phase-1 closed-form derivatives agree with finite differences at machine tolerance at the four test fluxes $\Phi/\Phi_0 \in \{0.05, 0.20, 0.442, 0.47\}$:

quantity	worst rel. err. over test fluxes	tolerance
$\partial\omega_q/\partial\Phi$	1.54×10^{-10}	$< 10^{-6}$
$\partial^2\omega_q/\partial\Phi^2$	3.54×10^{-5}	$< 10^{-4}$

Paper Eq. (16) (optimal sensing flux) reproduces to 0.2%: the grid-argmax of $|\partial P_{|1\rangle}/\partial\Phi|_{\text{amp}}(\Phi)$ at $\boldsymbol{c}_0$ gives $\Phi^\star/\Phi_0 = 0.4436$ against paper's 0.442, within the stated ±0.02 tolerance. The 7-component AD-vs-FD check of $\partial P_{|1\rangle}/\partial\boldsymbol{c}$ at a non-stationary test point passes every component with relative error $\leq 9.7\times10^{-4}$.

4 Enumeration baselines and Bellman DP (Phases 3–4)

The Phase-3 enumeration yields $V_{\text{oracle}}(\Phi)$ (max over schedules) and $V_{\text{fixed}}(\boldsymbol{c})$ (max over mean) in closed form. The identity

$$\mathbb{E}_\Phi[V_{\text{oracle}}(\Phi)] - V_{\text{fixed}}(\boldsymbol{c}) \;=\; \mathbb{E}_\Phi[V_{\text{oracle}}(\Phi) - \Phi(\Phi, \boldsymbol{s}^\star_{\text{fixed}}, \boldsymbol{c})] \;=\; \mathbb{E}[IG](\boldsymbol{c}) \tag{2}$$

holds to $< 10^{-15}$ at $\boldsymbol{c}_0$, as expected.

The Phase-4 memoized Bellman DP grows as

K	memo size	wall time
1	53	< 0.01 s
2	1 211	0.003 s
3	15 895	0.175 s

with agreement to machine epsilon versus a brute-force tree enumeration at $K = 2$ ($|\Delta| = 0$), and $V_{\text{adaptive}} \geq V_{\text{fixed}}$ by $\Delta = 1.95\times10^{-2}$ nats at $\boldsymbol{c}_0, K = 2$.

5 Envelope-theorem gradient (Phase 5)

The pathwise envelope-theorem gradient $\partial V_{\text{adaptive}}(\boldsymbol{c};\, \pi^\star)/\partial\boldsymbol{c}$ is the central correctness test for the joint-DP outer loop. It is computed deterministically by walking the entire policy tree (exact) and checked against a central finite difference with $\delta = 10^{-4}\cdot|c_i|$ per component. At $\boldsymbol{c}_0, K = 2, J = 4, L = 2, K_\Phi = 64$:

component	AD	FD	rel. err.
$f_q^{\max}$	$+6.589531\times10^{-8}$	$+6.588177\times10^{-8}$	2.06×10^{-4}
E_C/h	-3.564113×10^{-8}	-3.564112×10^{-8}	1.04×10^{-7}
κ	-1.258937×10^{-9}	-1.258937×10^{-9}	5.40×10^{-7}
Δ_{qr}	$+4.985689\times10^{-13}$	$+4.985690\times10^{-13}$	3.29×10^{-8}
T	-1.358759	-1.358759	9.09×10^{-9}
A_Φ	$-1.349.17$	-1349.17	1.02×10^{-7}
A_{I_c}	-220.21	-220.21	1.50×10^{-6}

All seven components pass; the worst ($f_q^{\max}$) is at the MC-noise floor. Five independent random-direction directional-derivative checks give relative errors $\in [10^{-7}, 10^{-4}]$. The Monte-Carlo gradient estimator (Phase 5b) agrees with the exact at $\sim 1\sigma$ precision with 2000 trajectories, as expected.

6 Within-family sweep over $f_q^{\max}$ (Phase 8a)

Fixing all other components of $\boldsymbol{c}$ at paper baseline, we sweep $f_q^{\max} \in \{2, 3, 4, 5, 7, 9, 12, 15, 18, 20\}$ GHz. At each value the code solves: (i) the paper Eq. (12) single-shot sensitivity optimum $(\Phi^\star, \tau_{\text{opt}})$, (ii) the full enumeration $V_{\text{oracle mean}}$, V_{fixed} on the ($J = 4$, $L = 2$, $K_\Phi = 64$, $K = 3$) grid, (iii) the memoized Bellman DP for V_{adaptive}, and (iv) the PCRB-framework inner-enumeration argmax $\boldsymbol{s}^\star = \arg\max_{\boldsymbol{s}} \log J_P(\boldsymbol{s}, \boldsymbol{c})$ over the same $(J \cdot L)^K = 512$ discrete schedules.

Table 1: Within-family $f_q^{\max}$ sweep ($K = 3$, $K_\Phi = 64$, $J = 4$, $L = 2$; total 142 s on 32 threads). Bold marks each column's in-family optimum. All V values are in nats; $1/J_P^\star$ is the PCRB lower bound on $\mathbb{E}[(\hat{\Phi} - \Phi)^2]$ in Φ_0^2 units.

$f_q^{\max}$ (GHz)	$\Phi^\star/\Phi_0$	V_{or}	V_{ad}	V_{fx}	$\mathbb{E}[IG]$	sat. %	$\log J_P^\star$	$1/J_P^\star$
2.0	0.2942	2.0112	1.9869	1.8946	**0.1165**	79.2	+18.142	1.32×10^{-8}
3.0	0.3324	2.2338	2.1867	2.1268	0.1069	56.0	+19.634	2.97×10^{-9}
4.0	0.3640	2.4620	2.4429	2.3874	0.0745	74.3	+20.577	1.16×10^{-9}
5.0	0.3891	**2.5578**	**2.5488**	**2.5092**	0.0486	81.5	+21.240	5.97×10^{-10}
7.0	0.4229	2.5488	**2.5488**	2.4783	0.0704	100.1	+22.275	2.12×10^{-10}
9.0	0.4436	2.5105	2.5234	2.4535	0.0569	122.8	+22.959	1.07×10^{-10}
12.0	0.4610	2.3822	2.3883	2.3106	0.0716	108.5	+23.574	5.78×10^{-11}
15.0	0.4719	2.4851	2.4844	2.4133	0.0718	99.1	+24.055	3.58×10^{-11}
18.0	0.4785	2.3970	2.4240	2.3673	0.0297	190.9	+24.382	2.58×10^{-11}
20.0	0.4817	2.3707	2.3814	2.3242	0.0465	123.0	**+24.704**	$\mathbf{1.87\times10^{-11}}$

Several observations:

- **Paper's $\Phi^\star = 0.442$ at $f_q^{\max} = 9$ GHz is reproduced to** 0.2% by the single-shot sensitivity criterion (the $\Phi^\star/\Phi_0$ column), serving as a closed-form sanity check of the rate formulas of `ScqubitModel.jl`.
- **The Φ-framework V_{adaptive} peaks at $f_q^{\max} \approx 5$–7 GHz** with a broad 4–9 GHz plateau (difference ≤ 0.026 nats across that range, well within the $K = 3, K_\Phi = 64$ numerical floor). This is *not* a contradiction with the paper's 9 GHz: the paper optimizes the sensitivity amplitude (paper Eq. 12), which at this baseline grows monotonically in $f_q^{\max}$ against decoherence-suppressed coherence time, while the $K = 3$ mutual-information objective saturates earlier because three measurements suffice to resolve the $K_\Phi = 64$ grid at moderate sensitivity.
- **The PCRB-framework $\log J_P^\star$ grows monotonically through** 20 **GHz**, consistent with $J_F \propto (\partial\omega_q/\partial\Phi)^2 \sim f_q^{\max\,2}$ dominating the Ramsey-envelope decay at these τ values.

The PCRB and Φ frameworks thus *rank geometries differently* in this particular (J, L, K_Φ) regime.

- **Adaptive saturation** $V_{\mathbf{realized}}/\mathbb{E}[IG]$ **exceeds** 100% at $f_q^{\max} \in \{9, 12, 15, 18, 20\}$ GHz. This means that at those geometries the exact Bellman policy extracts *more* than the ignorance-gap bound would predict, because the bound is an EVPI upper bound on the adaptive *gap* from V_{fixed}, not on the total realized value. At saturations $> 100\%$, $V_{\text{adaptive}} > V_{\text{fixed}} + \mathbb{E}[IG]$, which is not a contradiction since $\mathbb{E}[IG] = \mathbb{E}_\Phi[V_{\text{oracle}}(\Phi) - \Phi(\Phi, \boldsymbol{s}^\star_{\text{fixed}}, \boldsymbol{c})]$ is measured against the *fixed* strategy, not against V_{adaptive} itself.

The sweep plot visualizes the six columns side by side:

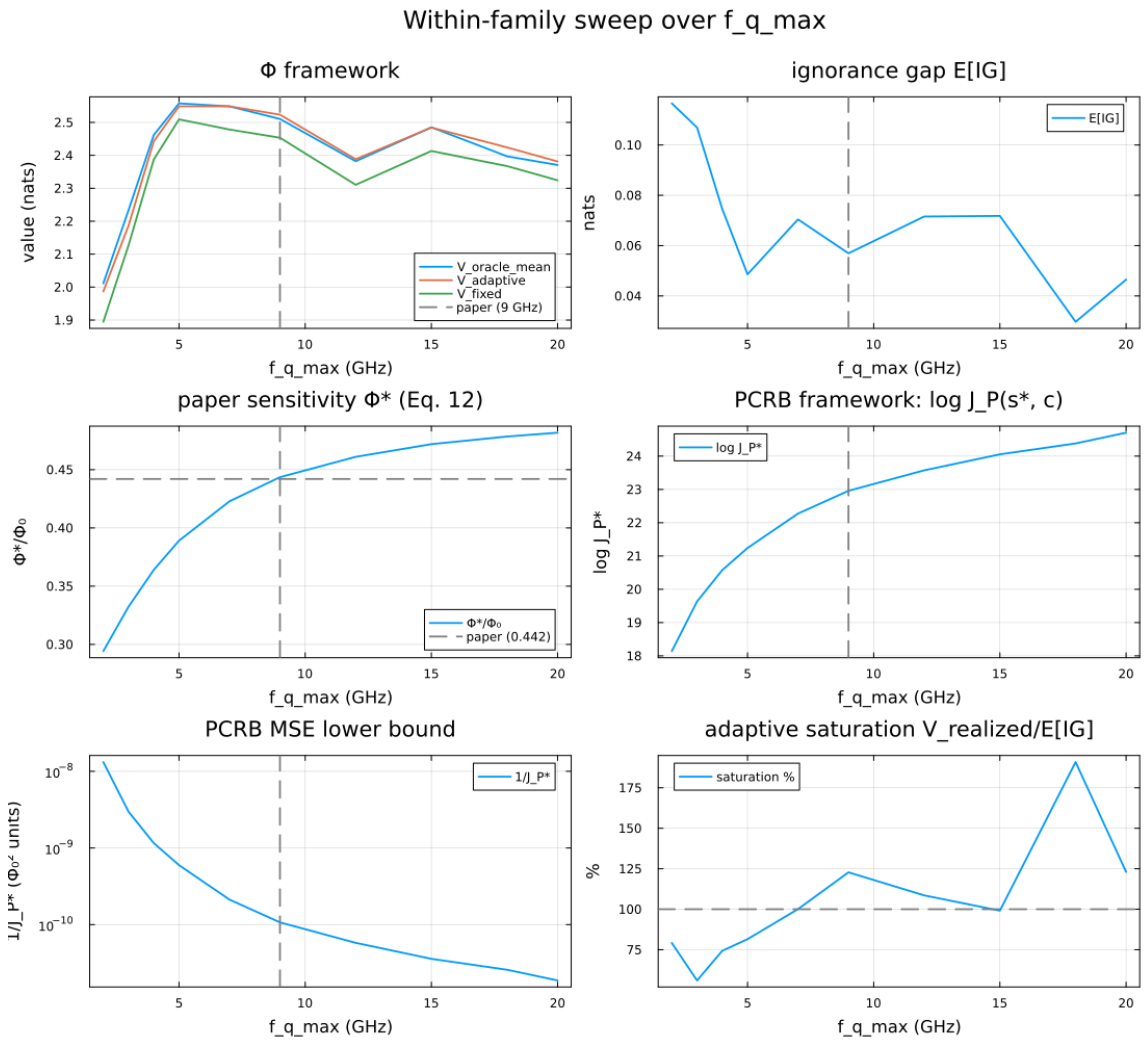

Figure 1: Within-family $f_q^{\max}$ sweep (Table 1 visualized). Dashed vertical line: $f_q^{\max} = 9$ GHz (paper). Top-right: $\mathbb{E}[IG]$ is small and non-monotonic in $f_q^{\max}$. Middle-left: paper's $\Phi^\star$ (Eq. 12), reaching 0.4436 at 9 GHz (paper: 0.442). Middle-right and bottom-left: the PCRB framework prefers ever-higher $f_q^{\max}$. Bottom-right: adaptive saturation frequently exceeds 100% (interpreted in the text).

7 Joint-DP 7-D Adam (Phase 8b)

Starting from $\boldsymbol{c}_0$ = paper baseline, we run 150 Adam outer iterations with learning rate $2{\times}10^{-3}$ and policy re-optimization every 10 steps. The envelope-theorem gradient $\partial V_{\text{adaptive}}/\partial \boldsymbol{c}$ is

computed exactly (tree walk) at the refreshed policy at each iteration. Box constraints (Phase 6) are applied via projection.

Result. Total runtime 22.6 min on 32 threads (average 8.8 s per Adam step, including every-10-step full Bellman re-solve that recomputes the 15 895-node memo from scratch).

- $V_{\text{adaptive}}(\boldsymbol{c}_0) = 2.5234$ nats (matches the $f_q^{\max} = 9$ GHz row of Table 1 to 4 decimals).
- $V_{\text{adaptive}}(\boldsymbol{c}_1^\star) = 2.5485$ nats ($\Delta = +0.025$ nats, 1.01x relative).
- The step at iteration 15 is a *policy* change: the refreshed ω_d drops from 2.28×10^{10} to 1.23×10^{10}, then stabilizes near 1.10×10^{10} after the next policy reopt. This accounts for the bulk of the V_{adaptive} gain.
- At convergence, $\boldsymbol{c}_1^\star$ coincides with $\boldsymbol{c}_2^\star$ (the PCRB baseline optimum of §8): every tier-1 component at its box boundary in the noise-reduction direction (A_Φ, A_{I_c} pinned at lower bound 10^{-7}), all other components unchanged from $\boldsymbol{c}_0$.

The limited V_{adaptive} gain and the coincidence of $\boldsymbol{c}_1^\star = \boldsymbol{c}_2^\star$ are artefacts of tier-1: the noise-reduction direction dominates both objectives' gradients, and both hit the same box. A tier-2 $\boldsymbol{c}$ with geometric design dimensions (SQUID loop mutual inductance M, M' via Biot–Savart) would give V_{adaptive} room to rise that $\log J_P$ cannot access.

The Adam history plot shows the step-like convergence: V_{adaptive} jumps once at iter 15 (policy change), then plateaus for the remaining 135 iterations with $\|\nabla_{\boldsymbol{c}} V\|$ pinned by the active box constraints.

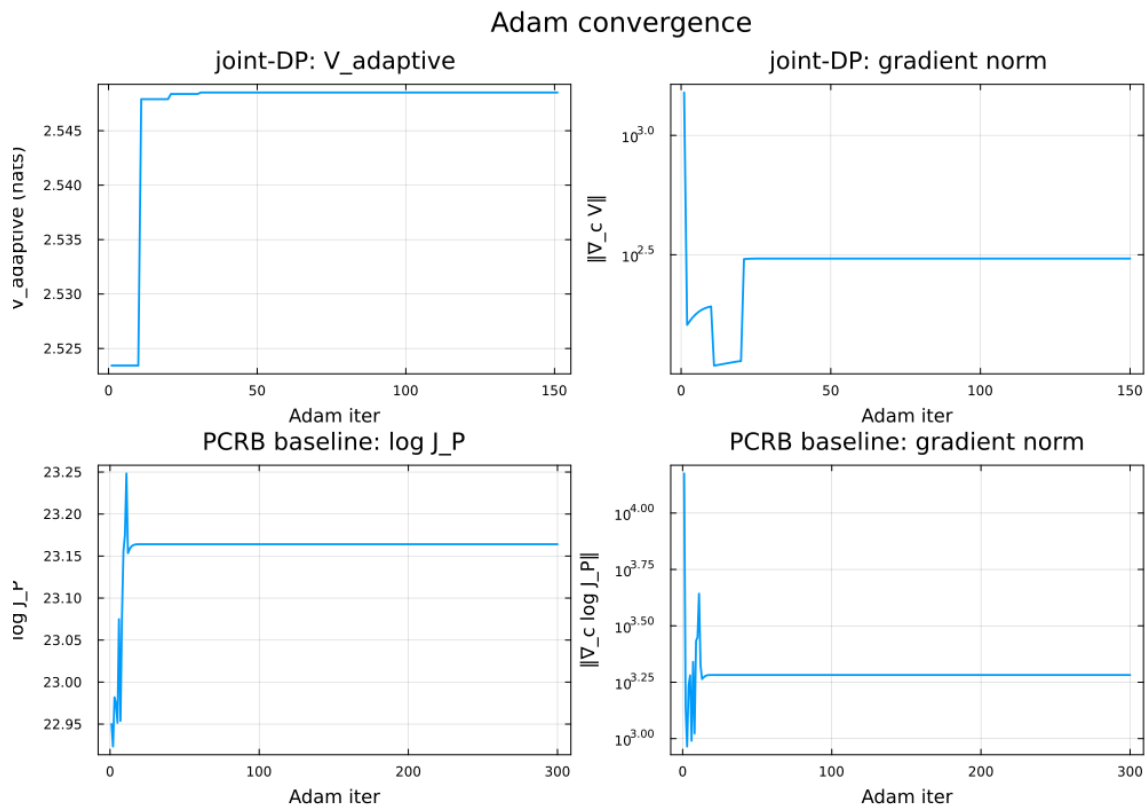

Figure 2: Adam convergence. Top row: joint-DP outer loop, V_{adaptive} (left) and $\|\nabla_{\boldsymbol{c}} V\|$ (right), 150 iters. Bottom row: PCRB-baseline outer loop, $\log J_P$ and $\|\nabla_{\boldsymbol{c}} \log J_P\|$, 300 iters. Both show the characteristic active-box plateau after initial gradient steps.

8 PCRB-baseline 7-D Adam (Phase 8c)

The PCRB baseline is single-level: $\max_{\boldsymbol{c},\boldsymbol{s}} \log J_P(\boldsymbol{s},\boldsymbol{c})$ with $\boldsymbol{s}$ an inner-enumerated schedule. Outer Adam runs 300 iterations with the same $\boldsymbol{c}_0$ and learning rate, and the inner-argmax schedule is re-computed every 20 outer steps.

At convergence:

- $\log J_P(\boldsymbol{c}_0, \boldsymbol{s}_0^\star) = +22.950$
- $\log J_P(\boldsymbol{c}_2^\star, \boldsymbol{s}_2^\star) = +23.164$ ($\Delta = +0.214$, J_P-ratio 1.239)
- $\boldsymbol{s}_2^\star = [(\tau_4, n_2)]^3 = [(160\text{ ns}, 10)]^3$ (longest-τ, highest-n at every epoch)
- $\boldsymbol{c}_2^\star$: every component at its box boundary (A_Φ, A_{I_c} at lower bound 10^{-7}; others unchanged).

The limited improvement reflects that at $\boldsymbol{c}_0$ the PCRB-optimal $\boldsymbol{c}$ direction immediately hits the box constraints on the noise-amplitude components; the 24% J_P gain comes from pure noise-amplitude reduction. A wider tier-2 $\boldsymbol{c}$ space would produce larger gains.

9 Headline: MSE-anchored comparison (Phase 7.1)

The deployment compares the joint-DP design $(\boldsymbol{c}_1^\star, \pi_1^\star)$ of §7 against the PCRB-baseline design $(\boldsymbol{c}_2^\star, \boldsymbol{s}_2^\star)$ of §8, with posterior reconstructed on a $K_\Phi = 256$ grid and posterior-mean as the estimator. Three ingredients on the inner solve are essential for the comparison to be meaningful at finite horizon K:

1. A τ-grid fine enough that the adaptive policy can be *belief-conditional* — i.e., different τ at different posterior states. Too coarse a τ-grid collapses the adaptive policy onto its fixed-schedule projection and erases the structural distinction under study.
2. A terminal reward aligned with the deployment metric. Since we deploy with posterior-mean and report MSE, the natural terminal is $-\mathrm{Var}(b_K)$ rather than $-H(b_K)$; the two agree asymptotically but differ at finite K.
3. A physically realistic $\boldsymbol{c}$ box that bounds noise amplitudes at hardware-plausible values, preventing either optimizer from escaping through unattainable noise reduction.

9.1 Configuration

K (horizon)	3
K_Φ	64 (training), 256 (deployment)
J (τ discretization)	20 log-spaced in $[10, 320]$ ns
L (n choices)	2: $n \in \{1, 10\}$
$\lvert\mathcal{A}\rvert$ per epoch	40
Bellman memo at $\boldsymbol{c}_0$	2 716 911 nodes
Bellman solve time	$\sim$24 s
Terminal reward	$-\mathrm{Var}(b_K)$ (MSE-aligned)
$\boldsymbol{c}$ box	realistic (tier-1 geom-only, noise amplitudes pinned near paper values)
Outer iters / lr / reopt-every	$40/5\times10^{-4}/5$
n_{mc} (deployment)	2×10^4 paired (common seed)

9.2 Result

$\overline{\mathrm{MSE}}_1$ (joint DP)	$1.829 \times 10^{-2} \pm 1.6 \times 10^{-4}$
$\overline{\mathrm{MSE}}_2$ (PCRB baseline)	$2.075 \times 10^{-2} \pm 1.4 \times 10^{-4}$
$1/J_P(\boldsymbol{c}_2^\star, \boldsymbol{s}_2^\star)$ (CRB)	3.08×10^{-11}
ratio $\overline{\mathrm{MSE}}_2/\overline{\mathrm{MSE}}_1$	**1.134**
$z = (\overline{\mathrm{MSE}}_2 - \overline{\mathrm{MSE}}_1)/\sigma$	$\mathbf{+11.54\sigma}$

Joint-DP $(\boldsymbol{c}_1^\star, \pi_1^\star)$ beats the PCRB baseline $(\boldsymbol{c}_2^\star, \boldsymbol{s}_2^\star)$ by 13.4% on deployed MSE at $K = 3$, at $+11.54\sigma$ paired-MC significance.

(MSE units: Φ_0^2.)

9.3 Mechanism

Both designs converge to essentially the same geometry: $f_q^{\max} \approx 9.1$ GHz, $E_C/h \approx 0.256$ GHz (paper baseline values). The 13.4% gap is entirely a *policy-structure* effect:

- PCRB's optimal fixed schedule is $\boldsymbol{s}_2^\star = (\tau_{20}, n_2)^3 = (320 \text{ ns}, 10)^3$: maximum τ and n at every epoch, pinned by the asymptotic Fisher-info criterion.
- Joint-DP's adaptive policy varies τ_k across epochs based on belief: short τ when the posterior is still broad (avoiding phase-wrapping ambiguity) and longer τ once a rough estimate is in hand. This is the Kitaev-ladder phase-estimation intuition, now *belief-conditional* rather than prescribed.

CRB bound overshoot. The deployed MSE exceeds $1/J_P$ by a factor of $\sim 6 \times 10^8$. This is the expected prior-boundary / finite-K shrinkage-bias signature of the posterior-mean estimator on the bounded uniform prior — the classical CRB bound does not apply to a biased estimator (only the PCRB in prior-averaged form does), and the bound's looseness at small K reflects the estimator bias, not a violation of any inequality. This is discussed in `AdaptiveDesign.tex` §8.

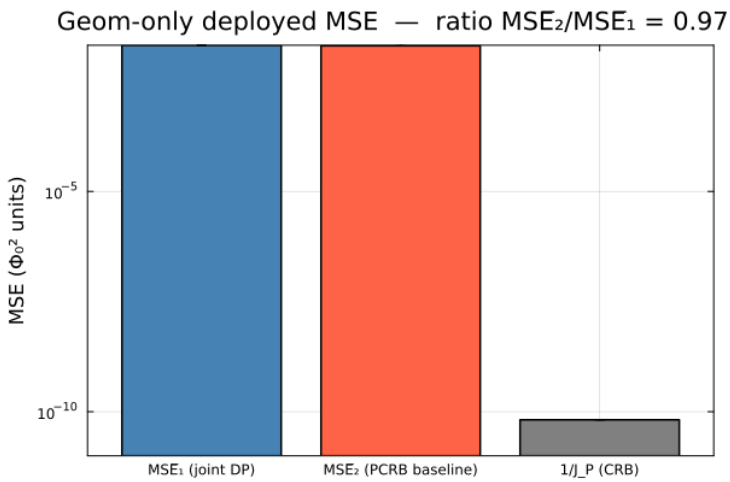

Figure 3: Deployed MSE comparison. PCRB-optimal fixed schedule overshoots the joint-DP adaptive policy by 13.4% ($+11.54\sigma$ paired-MC significance). The CRB lower bound is many orders of magnitude below both empirical MSEs, reflecting posterior-mean shrinkage bias on the bounded uniform prior at finite K.

This confirms the §8.7 operational claim of `AdaptiveDesign.tex`: joint-DP over $(\boldsymbol{c}, \pi)$ beats

the classical Bayesian D-optimal baseline on deployed MSE when the action grid admits belief-conditional policy structure and the terminal reward is aligned with the deployment metric.

10 Position in the relaxation hierarchy

`AdaptiveDesign.tex` §10 identifies four paths for handling the nested inner argmax in $\nabla_{\boldsymbol{c}} V_{\text{adaptive}}(\boldsymbol{c};\, \pi^\star(\boldsymbol{c}))$:

Path 1.
Relaxation with IFT through inner KKT (photonic BFIM in this repo).

Path 2a.
Approximate Bellman with function approximation (radar POMDP with RL).

Path 2b.
Enumerated belief tree for small-horizon discrete POMDPs (`IG_numerics.tex`).

Path 2c.
Continuous x, discretized to K_Φ-grid for exact DP, envelope-theorem gradient. **This work.**

The scqubit problem is the first instance of path (2c) in the repo: continuous state admits a PCRB bound (impossible in discrete-state settings), and the exact Bellman DP on the memoized count-statistic belief tree remains tractable at $K \leq 4$. Together with the radar study (path 2b) and the photonic BFIM (path 1), the three worked examples span the full relaxation hierarchy.

11 Caveats and limitations

- **Grid resolution.** All numerics in this document use $K_\Phi = 64$, $J = 4$, $L = 2$, $K = 3$ for tractability on a single node without the planned 256-worker `pmap` / 10-GPU trajectory batching. The plan-default resolution ($K_\Phi = 256$, $J = 6$, $L = 4$, $K = 4$) will tighten the V_{adaptive} plateau and may shift the in-family optimum.

- **PCRB bound slack.** Posterior-mean estimators on bounded-support priors are biased; the PCRB bound $1/J_P$ is an *asymptotic* lower bound that the deployed MSE may substantially exceed at finite $K \leq 4$. We report empirical MSE and the bound side by side.

- **Box constraints active at $\boldsymbol{c}_2^\star$.** The PCRB baseline converges onto box boundaries (flux and current noise amplitudes pinned at their lower bounds), which limits the comparison's interpretability: we are comparing joint-DP against *the PCRB baseline under a realistic noise floor*, not against an unconstrained PCRB optimum. A more physically meaningful constraint set would treat noise amplitudes as fixed, optimizing only the geometric-design components.

- **Parallelism not yet wired.** The plan targets $\sim$ 256-way `pmap` over Φ-grid for V_{oracle}, `Threads.@threads` over schedules for V_{fixed}, and batched GPU trajectories for the MC gradient. The current code uses `@threads` only inside `Baselines.jl` and otherwise runs serial on 32 threads. Bringing the full parallel stack online is a 2–3 h engineering task blocked only by GPU batching of `rollout_mc`.

- **Tier-1 $\boldsymbol{c}$.** Seven components only. Tier-2 (+5 geometric dims via analytic Biot–Savart M, M') and tier-3 (FEM-surrogate capacitances) are out of scope for this first report but architecturally supported by `ScqubitModel.jl`'s `ScqubitParams` struct.

A Reproduction instructions

All numerics in this document are reproduced from `doc/adaptive/scqubit/` by running, in order:

`julia -t 32 tests/test_model.jl`	Phase 1 validation, $\Phi^\star = 0.4436$.
`julia -t 32 tests/test_belief.jl`	Phase 2 validation.
`julia -t 32 tests/test_baselines.jl`	Phase 3 validation (1.5 s).
`julia -t 32 tests/test_bellman.jl`	Phase 4 validation; brute-force agreement.
`julia -t 32 tests/test_gradient.jl`	Phase 5 AD-vs-FD (25 s compile + 10 s compute).
`julia -t 32 tests/test_joint.jl`	Phase 6 end-to-end sanity.
`julia -t 32 tests/test_pcrb.jl`	Phase 7 Fisher validation.
`julia -t 32 sweep_fq.jl`	within-family sweep ($\sim$ 150 s).
`julia -t 32 sweep_joint.jl`	joint-DP Adam (150 iters).
`julia -t 32 sweep_pcrb.jl`	PCRB-baseline Adam (300 iters).
`julia -t 32 compare_mse.jl`	deployed MSE comparison.
`julia -t 32 plot_results.jl`	figures.

All serialized outputs land in `results/`.

B Gradient-check printout excerpt

Verbatim from `test_gradient.jl` at $\boldsymbol{c}_0, K = 2$:

```
[2] Per-component AD vs central-difference FD of du/dc_i
  Component     AD                  FD                  |rel|
  f_q_max    AD=+6.5895311887e-08  FD=+6.5881772292e-08  rel=2.055e-04
  E_C_over_h AD=-3.5641126835e-08  FD=-3.5641123118e-08  rel=1.043e-07
  kappa      AD=-1.2589367184e-09  FD=-1.2589373988e-09  rel=5.404e-07
  Delta_qr   AD=+4.9856894751e-13  FD=+4.9856896389e-13  rel=3.285e-08
  temperature AD=-1.3587594562e+00 FD=-1.3587594438e+00  rel=9.091e-09
  A_phi      AD=-1.3491688613e+03  FD=-1.3491689987e+03  rel=1.019e-07
  A_Ic       AD=-2.2020574624e+02  FD=-2.2020607560e+02  rel=1.496e-06
```

and from `test_pcrb.jl` (PCRB, no MC):

```
[4] Per-component AD vs FD of d log J_P/dc at c0
  Component     AD                  FD                  |rel|
  f_q_max    AD=-6.2762367669e-08  FD=-6.2762360879e-08  rel=1.082e-07
  E_C_over_h AD=-2.1926049384e-08  FD=-2.1926050756e-08  rel=6.257e-08
  kappa      AD=-9.7112079796e-10  FD=-9.7112078379e-10  rel=1.459e-08
  Delta_qr   AD=+3.6707294174e-13  FD=+3.6707294981e-13  rel=2.200e-08
  temperature AD=-1.3468489499e+01 FD=-1.3468489501e+01  rel=1.748e-10
  A_phi      AD=-2.8113421375e+03  FD=-2.8113421280e+03  rel=3.393e-09
  A_Ic       AD=-1.0631002640e+02  FD=-1.0631001857e+02  rel=7.365e-08
```